Total Pages 36 (Thirty Si

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SAVITRIBAI PHULE PUNE UNIVERSITY

Seat No S50022024
Stk. No 3497249
Sub. DS
Centre 4022

PCC-201-COM-DS

Sem:3 7022416

Examination	END-SEM (SPP Data Structure
Day & Date	15 Dec 25, Monda
Subject	Data structure
Paper No.	65827-236 Sec
Medium of Answer	English
Seat No. : In figure & In word	S 5 0 0 2 2 0 2 4 S five zero zero two two zero two four seven
Signature of Candidate	
Signature of Supervisor	V. 15/12/202
Center No.	4 0 2 2

Instruction to Candidate

1. Candidate has to confirm seat number, subject and centre number printed on Bar code and write it on attendance sheet.
   विद्यार्थ्याने प्रथम बार कोडवरील आसन क्रमांक, विषय व केंद्र क्रमांक तपासून योग्य असल्याची खात्री करावी आणि उपस्थिती पत्रकावर नोंदवावी.
2. Paste Bar Code in prescribed space.
   उत्तरपत्रिकेवरील विहित जागेतच बार कोड लावावा.
3. Do not write anything on Bar code sticker, otherwise it will be treated as unfairmeans.
   बार कोड स्टिकरवर काहीही लिहू नये, अन्यथा परीक्षा गैरप्रकार समजला जाईल.

Q. No.	Examiner	Moderator
1		
2	08	
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Total in Figure	46	
Total in Words	Four Six	
Signature		

Total	Marks in Figure	Marks in Words	Sign
Examiner	46	Four Six	
Moderator			

१. विद्यार्थ्याने उत्तरपत्रिकेच्या मुखपृष्ठावर तसेच उपस्थिती पत्रकावर विहित जागेत आसन क्रमांक अंकात व अक्षरात बिनचूक लिहून स्वाक्षरी करावी.

२. उत्तरपत्रिकेवर फक्त निळ्या अथवा काळ्या शाईचा उपयोग करावा, अन्यथा उत्तरपत्रिकेचे मूल्यमापन केले जाणार नाही.

३. उत्तरपत्रिकेच्या पृष्ठक्रमांक ३ पासून लिहिण्यास प्रारंभ करावा.

४. संबंधित प्रश्नाचे अथवा उपप्रश्नाचे उत्तर जेथून सुरू होते तेथेच समासात प्रश्न क्रमांक, उपप्रश्न क्रमांक अचूक व स्पष्ट लिहावा, यासाठी वेगळ्या शाईचा उपयोग करू नये.

५. प्रत्येक पानाच्या दोन्ही बाजूस लिहावे, उत्तरपत्रिका किंवा पुरवणी उत्तरपत्रिकेचे कोणतेही पान फाडू नये, फाडल्यास परीक्षा गैरप्रकार समजून पुढील कार्यवाही करण्यात येईल.

६. पेपर संपण्यापूर्वी १० मिनिटे अगोदर इशारा घंटा होईल, त्यानंतर विद्यार्थ्याने उत्तरपत्रिका व पुरवणी उत्तरपत्रिकेवर होलोक्राफ्ट स्टिकर विहित जागेवरच लावावा.

७. कॉपी करणे किंवा दुसऱ्याच्या नावावर परीक्षेस बसणे यांसारख्या कृती 'महाराष्ट्र-प्रीव्हेन्शन ऑफ मालप्रॅक्टिस ॲक्ट युनिव्हर्सिटी, बोर्ड ॲण्ड अदर स्पेसिफाईड एक्झामिनेशन्स ॲक्ट, १९८२' (सा.फु.पु.वि. चा अध्यादेश क्रमांक ९) त्यानुसार संमत केलेला कायदा या अन्वये दंडही असेल.

Candidate shall fill all information about se number, paper etc. in prescribed space and sign the answer book and attendance sheet.

Candidate shall use blue or black ink onl Otherwise answer book will not be evaluated.

Candidate shall start writing answer from page n 3 of the answer book.

Candidate shall mention question number, su question number correctly at the beginning of th same and shall not use ink other than blue or blac

Candidate shall write on both sides of pages an shall not tear off any page. it will be treated unfairmeans.

Warning bell will be given before 10 minutes of th concluding time. Candidate shall paste Holocra Sticker at appropriate space on the answer book

An Act of Copying or Impersonations at Examination is Punishable under 'The Maharasht Prevention of Malpractices at University, Board an Other Specified Examinations Act, 1982' (Ordinan 9 of SPPU). The Act passed to the effect.

Examiner and Moderator has to write marks all given appropriate place only. Examiner shou give assessment tick(✓) or (×) in the margin.

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Total	46			

Q2.

Static Data Structure	Dynamic Data Structure
1. Static Data Structure are those data structure that are formed during completion time.	1. Dynamic Data structure are those data Structure that are formed during runtime or execution time.
2. The size of the static Data structure is fixed	2. The size of the Dynamic Data Structure is not fixed means it can change according to the given data.
3. The Data at accessing or removing is fast because of fixed size	3. The Data accessing or removing is slow because of dynamic size.
4. Unwanted data is store or unrequired things are there because of fixed size	4. No-unwanted data is stored because of its dynamic size
5. eg. Array size of Array is fixed eg. 3 4 5 6 7 size of array is 6 [3 | 4 | 5 | 6 | 7 |] empty	5. eg. Linked list size of linked list is not fixed it depend on data eg. 3 4 5 6 7 [3 | 4 | 5 | 6 | 7]

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Persistent Data structure :

In this type of Data structure we not only get access to the recent data but also to the previous data.

- Fully Persistent Data : It get access to complete recent data and previous data

- Partially Persistent Data : In it only partial data can be accessed.

Ephemeral Data Structure

In this type of Data structure we only get access to the recent data only

Means we will only able to get access of the recent data.
We cannot get access of the past data

b. Step count method : It is a method in which we count / count time at every step and then add the total time.

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Code	step count	Variable
def oddEven(n=100):	0	c_1
for num in range(1,n):	n	c_2
if num%2==0	n-1	c_3
print(num,'is Even')	n/2	c_4
else:	0	c_5
print(num,'is odd')	n/2	c_6

∴ Total no of step count : $c_1+c_2+c_3+c_4+c_5+c_6$

$0+n+n-1+n/2+n/2$

$\boxed{3n-1}$

∴ step count is $3n-1$

⇒ The above algorithm has linear time complexity because it has only 1 for loop and power of n is 1

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Algorithm strategies

Algorithm strategies are those strategies are that we use to get time complexity.

1. $O(1)$
2. $O(n)$
3. $O(n^2)$
4. $O(\log n)$
5. $O(n \log n)$
6. $O(n!)$
7. $O(2^n)$

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Q3-

1 Sparse Matrix : A Sparse Matrix is a type of Matrix which has more zero than the non-zero element

eg.

$$\begin{array}{c|ccccc} & 0 & 1 & 2 & 3 & 4 \\ \hline 0 & 1 & 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 1 & 0 & 0 \\ 3 & 0 & 1 & 0 & 0 & 1 \\ 4 & 0 & 0 & 0 & 1 & 1 \end{array}_{m \times m}$$

The given matrix is the Sparse Matrix in which has more zeros than non zero

It is 8 non-zero element

It has 17 zero element

Row	Colum	Element (non zero)
0	(1, 2,)	1, 2
1	2	1
2	2	1
3	1, 4	1, 1
4	3, 4	1, 1

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2. <u>Sorting Stability</u> : In it the order of 2 same data/element will be same after the sorting

or

The order remain same of 2 same element at Input and output after sorting. eg. bubble, selection

eg. element [2a 1 5 2b 4] → This is Input
index 0 1 2 3 4

a and b are same

After sorting

[1 2a 2b 4 5]
↓ Sorted ↓ Stable

So even after sorting the order of same element remain same so it is sorting stability.

3. Internal and External

In <u>Internal sort</u> the sorting takes place inside the main memory or in the RAM (Random Access Memory)

- This type of sorting is done for <u>small data</u>
- It is <u>faster than</u> external sort

eg. Quick sort, Merge sort, bubble sort etc.

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External sort : It is done for large amount of data

- It is slower than Internal sort
- It is not done inside the main memory.

eg. Extended merge sort.

b) Binary search :

Given : 9, 17 23, 38, 45, 50, 57, 79, 90, 10

To find : 23, 90

For 23

element	9	17	23	38	45	50	57	79	90	100
index	0	1	2	3	4	5	6	7	8	9

Step 1: left = 0 , Right = 9

$$Mid = \frac{left + Right}{2} = \frac{0+9}{2} = 4.5 \Rightarrow 4$$

T.C at this step $\Rightarrow \frac{n}{2}$

Step 2: index 4

element is 45

45 > 23

$\Rightarrow$ Now Right = Mid − 1

= 4 − 1 = 3

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Mid = $\frac{0+3}{2} = 1.5 \Rightarrow 1$

This step $\left(\frac{\frac{n}{2}}{2}\right) \Rightarrow \frac{n}{4}$

At index 1 ⇒ 17

23 > 17

Step: left = mid + 1
= 2

Mid = $\frac{2+3}{2} = \frac{5}{2} \Rightarrow 2$

$\left(\frac{\frac{n}{4}}{2}\right) = \frac{n}{8}$

At index 2 ⇒ 23

23 = 23

Element found at index 2

⇒ Find 90

	9	17	23	38	45	50	57	76	90	100
Index	0	1	2	3	4	5	6	7	8	9

left = 0 , right = 9 mid = $\frac{\text{left} + \text{right}}{2}$

Step 1: mid = $\frac{0+9}{2} = 4.5 \Rightarrow 4$

At index 4 ⇒ 45

$\boxed{90 > 45}$

$\frac{n}{2}$

$\boxed{\text{left} = \text{mid} + 1}$ left = 4 + 1 ⇒ 5 , right = 9

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Step 2 : mid $= \frac{5+9}{2} = \frac{14}{2} = 7$ $\left(\frac{\frac{n}{2}}{2}\right) \Rightarrow \frac{n}{4}$

At index 7 $\Rightarrow 76$

$90 > 76$

left = mid + 1 , left = 7 + 1 $\Rightarrow 8$

Step 3 : mid $= \frac{8+9}{2} = 8$ $\left(\frac{\frac{n}{2}}{4}\right) = \frac{n}{8}$

At index 8

$\underline{90 = 90}$

Found at index $\underline{8}$

$\Rightarrow$ T.C at each search

T.C of Binary search is $O(\log(n))$

$\Rightarrow O\left(\frac{n}{2}\right) + 1$

$\frac{n}{4}\left(\frac{\frac{n}{2}+1}{2}\right) + 1$

$\Rightarrow O(\log(n))$

At k position it is $\frac{n}{2^k}$

Let $\frac{n}{2^k} = 1$

$n = 2^k$

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$\log n = K \log_2 2$

$T.C = K = O(\log n)$

For 23, index found is 2

$\therefore \quad T.C = O(\log n)$

$= \log_2 2$

$\boxed{T.C = 1}$

$T.C = O(\log n)$

$\boxed{T.C = \log_{10} 2}$

For 90, index found is 8

$T.C = O(\log n)$

$= \log_2 8 \Rightarrow \log_2 2^3$

$= 3 \log_2 2$

$\boxed{T.C = 3}$

$T.C = O(\log n)$

$= \log_{10} 8 = \log_{10} 2^3$

$= 3 \log_{10} 2$

$\boxed{T.C = 3 \log_{10} 2}$

E. /Q.No.

Q6 (a)

① Queue : ~~Queue is a linear data structure. It is used to store element in.~~

Q5.

(a)

1. SLL (Single linked list) : It is a type of linked that is unidirectional
   - The last node point to null
   - The Drawback of it is if the data from a node is deleted then it will remain empty

Data	Next →

- It only has next pointer

eg. 23 → 28 → 39 → 40 → null

- last node point to the null
- It has only next pointer

Like for 23 point towards 28,
28 towards 39
39 towards 40
40 towards null

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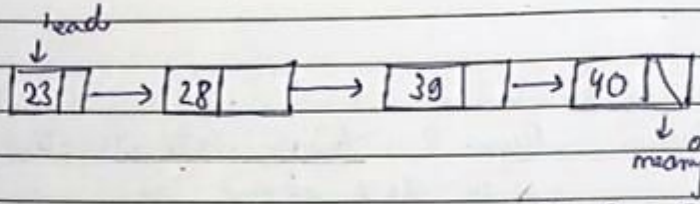


Pointer is point to null

2. DLL (Double linked list) : In this linked list we can move in 2 direction or pointers point towards 2 Direction

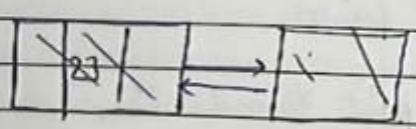
Prev	Data	Next

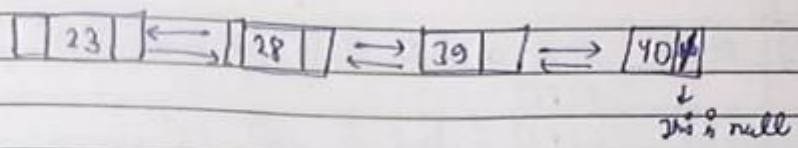
- Data Point towards 2 direct / Pointer are
- Prev ; Next

- Prev point towards the previous data
- Next point towards the next data
- /

eg. 23 28 39 40

23 ⇄ 28 ⇄ 39 ⇄ 40

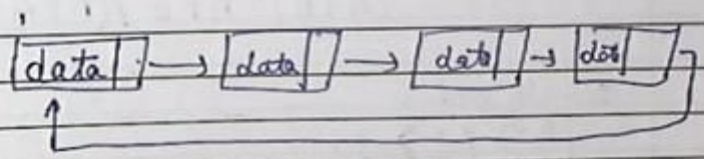




- It is usefull as we can move in 2 direction
- The last node next pointer points towards the null.

(iii) CLL (Circular Linked list) : In circular linked list there is no null. The no

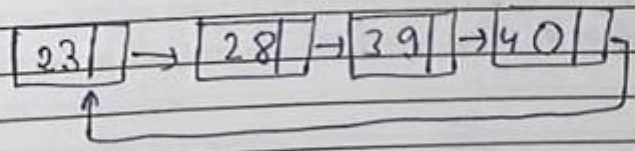
- The last node point towards the first node
- It is good because we can go back to data again and again.
- It is unidirectional



Last node point towards to first node / head.

- Never a null point.

eg.



There is no null point.

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(b) Stack : It is a linear data structure that follow LIFO
- LIFO is last in first out.

Stack as an ADT → ADT is Absolute data type in stack
- It just tell what operations to follow but do not tell about the implementation part

- It do not tell how to implement.

In stack it tell Insertion, deletion, pop, push, peek
but will not tell about their implementation.

(ii) infix expression
$(A+B)*D+E/(F+A*D)+C$

Priority order
$\boxed{\wedge > / * > +, -}$

- Create a empty stack
- place operator in stack
- Operands in output.

Q.No.

expression.

(A+B)*D+E/(F+A*D)+C

Inputs	stack	Output /Result
(	[( ]	
A	[( ]	A
B +	[(+]	A
B	[(+]	AB
)	[ ]	AB+
*	[*,]	AB+
D	[*,]	AB+D
+	[+]	AB+D*
E	[+]	AB+D*E
/	[+, /]	AB+D*E
(	[+, /, ( ]	AB+D*E
F	[+, /, ( ]	AB+D*EF
+	[+, /, (, +]	AB+D*EF
A	[+, /, (, +]	AB+D*EFA
*	[+, /, (, +, *]	AB+D*EFA
D	[+, /, (, +, *]	AB+D*EFAD
)	[+, /,]	AB+D*EFAD*+/
+	[+, +]	AB+D*EFAD*+/+
C	[+]	AB+D*EFAD*+/+C
		AB+D*EFAD*+/+C+

Postfix. AB+D*EFAD*+/+C+

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∴ Postfix is AB + D* EFAD* +/+C+

Q7.

(a) Good Hash function

1. Low collision : In for Good Hash function their it should be Low collision.
Means:
A Hash function should be made like this that there should be low no of collision in hash-table.

2. Even distribution : For Good Hash function there should be Even distribution of elements in the Hash-table.

• Mean. A hash function should make it possible so that elements are evenly distributed in Hash table.

3. Low Clustering : A Good Hash function there should be low clustring

• Means low accumulation of elements at a place.

4. Hash table is not ever overflowing.

(ii) Four type of hash function are

1. Multiplication hash function
2. Division hash function
3. Mid square hash function
4. Folding hash function

1. Division Hashing fution : In this hash function element is modulus & taken with the size of table.

eg. Size of table = K

Let elmt is = m

In Hash table its index is ~~K%m~~ m % K

eg. 32, ~~40~~, 51, 53

size is 4

index = 32 % 4
= 0

index = 51 % 4
=

index	
0	32
1	
2	
3	51

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2. Mid square folding method : In Mid square folding method we first do square of element and then take the middle elmt of that elmt as index

eg. 12 → 12^2
= 121

Middle elmt is 2.
It's index is 2

3. Folding method : In As the name suggest in this we group the elmt and add the elmt and take the middle part

$\underbrace{100}_{1}\underbrace{214}_{2}\underbrace{621}_{3}$ 16

100
214
621
935.

It size of table is 10, index is 3

4. Multiplication method : In this method we take a elmt and k from $0<k<1$

and multiple it to A $\sqrt{n} - 1$

Index = (Elem + KA)

(b)

Given:

modulo : 10

Elements : 24, 13, 16, 15, 19, 20, 22, 14, 17, 26

Index = Elements % size of table.

0	
1	
2	
3	13
4	24
5	15
6	16
7	
8	
9	19

(1) 24 % 10
Index = 4

(2) 13 %.
13 % 10
index = 3

3. 16
16 % 10
Index = 6

4. 15
15 % 10
index = 5

5. 19
19 % 10
Index 9

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~~Index~~

5. 20
20%10
Index = 0

7. 22
20%10
Index = 2

8. 14
14%10
index = 4
Collision occur at 4

20	0
	1
22	2
13	3
24	4 ← collision
15	5
16	6
	7
	8
19	9

Move to next empty place

20	0
	1
22	2
13	3
24	4
15	5
16	6
14	7
17	8
19	9

9. 17
17%10
index = 7
collision

Move to next empty place

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10. 26

26%10

index = 6

collision.

move to next empty place.

0	20	
1	26	8
2	22	7
3	13	6
4	24	5
5	15	4
6	16	3
7	14	2
8	17	1
9	19	0

Average search time

1. 24 → 1
2. 13 → 1
3. 16 → 1
4. 15 → 1
5. 19 → 1
6. 20 → 1
7. 22 → 1
8. 14 → not at index for 4 time ∴ 4
9. 17 → not at index 7 time = 2.
10. 26 → not at index 6 time = 6

Total = ~~1+1+1+1+1~~
1+1+1+1+1+1+1
4+2+6
~~12~~

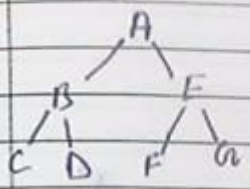
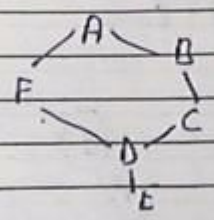
Total = 19

Average = 19/10 = 1.9

∴ Average search time is 1.9.

Q.9.

Tree	Graph
1. It is a non-linear data structure	1. It is a non-linear data structure
2. It is non-cycle, means not closed	2. It is cycle, means closed.
3. It is a subport of Graph	3. It is not the subport of Graph
4. All trees are graph	4. All ~~trees~~ Graph, are not trees.
5. In tree there is a root node, parents node, child node	5. In the Graph there is no such thing exist
6. Tree are the subport of graph	6. Graphs are not subport of graph
• Real life applicate of Tree's are	• Real life applicate of Graph are
• MST (Minimum spanning tree)	
• BST (Binary search tree	
Like In Photo album	

Diagram of Tree	Diagram of Graph
	
• It is divided like complete tree, full tree	• These are various graph complete graph, full graph
• Used in BST (Binary search tree)	• It is used to get MST (Minimum spanning tree
• A spanning tree is a part of tree that is used to get minimum cost of a thing.	• If a graph has n node/vertices then the number of spanning tree is n^{n-2}

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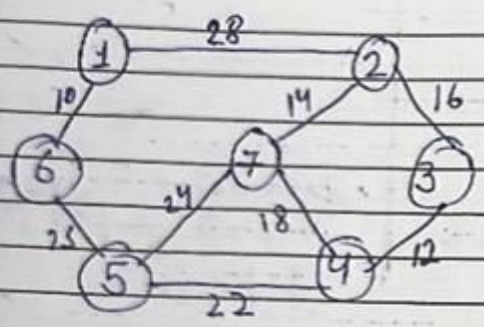
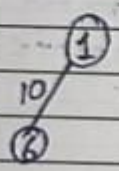
Q. K. /Q.No.

(b) Prim's.



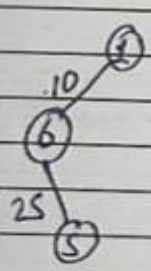
Given graph.

Step 1: Take any nodes



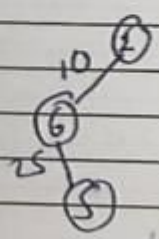
Vertex = (1)
Edges (28,10)
min (28,10)

Step 2.

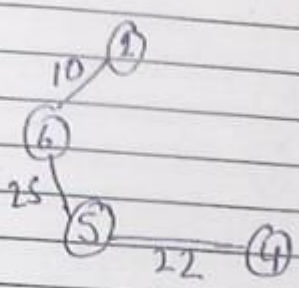


Vertex (1,6)
Edges (28,25)
min (28,25)

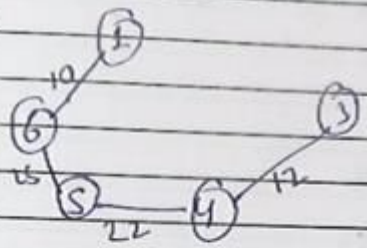
Step 3.



Vertex (1,6,5)
Edge (28,24,22)
min (28,24,22) = 22



Step 4



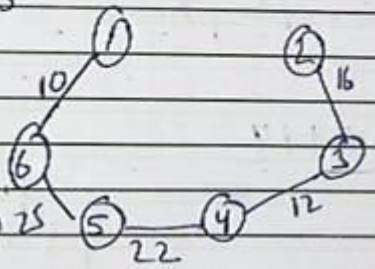
Vertex (1, 5, 4)

Edges (28, 24, 18, 12)

min

12

Step 5.

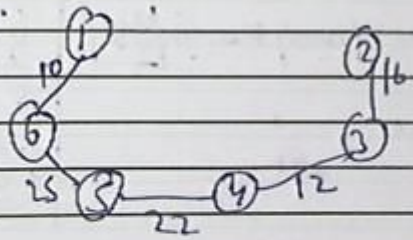


Vertex (1, 4, 3)

Edges (28, 18, 16)

min 16

Step 6.



Vertex (1, 5, 4, 2)

Edge (28, 24, 18, 14)

min.

14

SPPU- 28/ 36

	Q.No.						TOTAL
E							
M							

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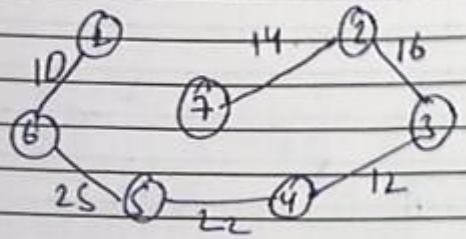
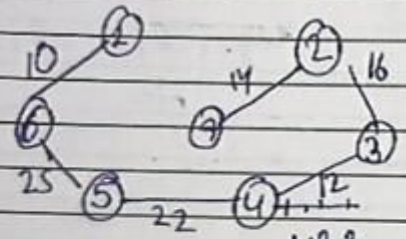


All vertex are complete this is Prim's Minimum spanning tree.



Minimum spanning tree.

cost is.

$10 + 25 + 22 + 12 + 16 + 14$

∴ cost is 99

6

∴ The cost of Minimum spanning tree is $\boxed{99}$

16
14
30
40
25
65
22
87
12
99

Q.No.

SPPU- 30/ 36

	Q.No.						TOTAL
E							
M							

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Q.No.

SPPU- 32/ 35

	Q.No.						TOTAL
E							
M							

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प्र. क्र./Q.No.

S. F. /Q.No.

SPPU-34/35

	Q.No.						TOTAL
E							
M							

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प्र. क्र./Q.No.

Q.No.

SPPU-36/36

	Q.No.						TOTAL
E							
M							

Savitribai Phule Pune University

प्र. क्र./Q.No.